



3rd May 2024

Dear Hiring Manager,

I am writing to express my interest in the **[Position Title]** role at your company. I came across this opportunity through **[source]** and am excited about the prospect of joining your team.

My journey began when I joined a worldwide Discord server with the goal of increasing member activity and retention while fostering a friendly global environment. As a software developer, I took on building a Discord bot using **Python**, despite being new to the language and development itself.

Rather than getting stuck in tutorials, I dove straight into the project, starting with basic features and gradually adding more complex functionalities like an economy system where members earned virtual currency for each message. This required data persistence, so I initially used a **raw JSON file** but quickly realized the limitations due to concurrent writes crashing the bot frequently.

Undeterred, I learned about databases, starting with **MongoDB's sharded cluster** for fast writes. As the project grew, I optimized performance by introducing **Redis caching** to address latency issues. When a relational structure was needed, I transitioned to **PostgreSQL** as the main database while utilizing Redis for caching.

Driven by curiosity, I explored **Rust** and built a **Data Access Layer using gRPC** for efficient communication with the Python bot. I integrated APIs like Axum to enable real-money currency purchases. To further optimize, I utilized **libSQL**, an embedded SQL database, within the Rust process to reduce network latency.

Key accomplishments include:

- Developed and scaled a Discord bot using Python, Rust, gRPC, MongoDB, Redis, PostgreSQL, SQLite, Axum
- Implemented an economy system, caching, database migration and embedded database integration
- Hosted on various platforms like repl.it, DigitalOcean, AWS, planning Next.js web dashboard on Cloudflare Workers

Throughout this journey, I demonstrated passion for learning, problem-solving abilities and consistently found innovative solutions. I am confident my technical expertise and intellectual curiosity make me an ideal fit for this role.

I am excited about the opportunity to join **[Company Name]** and contribute to your team's success. I look forward to discussing my candidacy further.

Sincerely,

Sudip Roy



29 June, 2024

Dear CEO of Oowlish,

I am writing to express my strong interest in joining **Oowlish** as a **Full Stack/Backend Software Developer**. Your company's innovative approach to providing dedicated tech squads and on-demand software development services aligns perfectly with my technical expertise and career aspirations.

My journey began when I joined a worldwide Discord server with the goal of increasing member activity and retention while fostering a friendly global environment. As a software developer, I took on building a Discord bot using **Python**, despite being new to the language and development itself.

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Throughout this project, I've demonstrated a commitment to continuous learning and innovation. I've tackled challenges such as concurrent write issues, optimized performance through caching and database selection, and integrated new technologies like Rust and gRPC to enhance system capabilities. I'm particularly excited about the opportunity to contribute to Oowlish's tech squads and on-demand development services. My experience in rapidly adapting to new technologies and solving complex problems would be valuable in delivering innovative solutions to your clients. As Oowlish's CTO, I imagine you're looking for developers who can not only code efficiently but also understand system architecture and make sound technical decisions. My journey from a novice programmer to architecting a complex, scalable system demonstrates my capacity for growth and leadership in technical roles. I am eager to bring my technical skills, problem-solving abilities, and passion for innovation to Oowlish. I look forward to the opportunity to discuss how I can contribute to your team's technical excellence and help drive Oowlish's mission forward. Thank you for your consideration.

Sincerely,

Sudip Roy